UNIT 2: SOIL AS ARCHIVE

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How does soil speak? What are its silences? In this unit we consider the histories that are embedded in, revealed by and erased by soil.

Michel-Rolph Trouillot writes:

> Silences enter the process of historical production at four crucial moments: the moment of fact creation (the making of *sources*); the moment of fact assembly (the making of *archives*); the moment of fact retrieval (the making of *narratives*); and the moment of introspective significance (the making of *history* in the final instance) (pp. 26–27)

This unit builds on the previous unit's exploration of soil's ability to hold and reflect historical metabolisms to consider its role in holding, reflecting or obscuring history more broadly. We ask the following questions to frame this unit's activities and readings:

- How can we envision soil as an archive – what does this help us see? What silences are embedded in this archive?
- How have soil archives been created and used and in the service of what projects? What erasures are embedded in this process?
- How can we reimagine soil as an archive – one that materialises the lasting legacies of imperial histories and racialised social relations?

Understanding how social and historical processes manifest materially in the soil can both help us to understand how we live with the legacies of past and ongoing processes, and to take control of the archive – to better understand what those legacies are and how we might reshape them.

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Day 1: Archives of soil

Today we continue some of the work we did in the prior unit on building our skills of assessment and judgement of soil while also considering what values such practices reflect and reproduce. These activities and readings aim to replicate and interrogate how scientists create and collect soil monoliths.



Lesson argument: Soil collections are archives in the sense that they bring together materials and organise them; in the sense that they further certain (violent) classification projects; in the sense that they tell historical stories; and in the sense that they have afterlives.

Questions

How have soil archives been created and used and in the service of what projects?

What are the implications of the typologies that have been created for understanding and archiving soil?

Activity: Creating soil monoliths

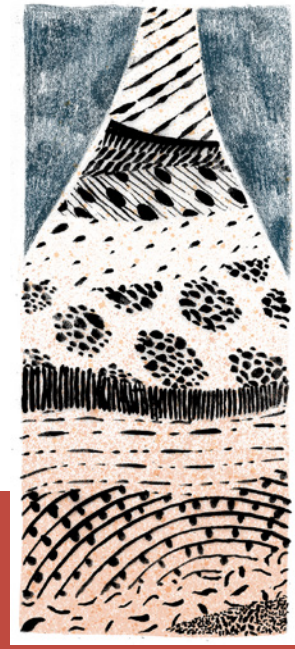
Find two sites near you where there is soil – in each, take a sample of the top soil using a clear cup. Compare them and try to describe the features you are comparing. What do the layers look like? If you don't see any layers, add dish detergent and water and shake it up, then wait for the sediments to fall. Look at the ratio of different layers: soil, silt, sand, clay.

Questions

Are you comparing based on colour? Based on texture? How would you make a classification system? To see one of these collections visit: https://wsm.isric.org/WSRC.html

Scientists have collections of these monoliths. Is making a monolith of your own considered an archive? If so, what is it archiving? What makes something an archive?

Reference:
Mounting-your-soil-monolith-Oct-2020_MW_Generic-for-MSSS-website22163.pdf



Assignment: Explore FAO/UNESCO Soil Map and discuss with questions

Look at the legend – consider the note on nomenclature on page 2. What do these naming choices suggest about the goals of this map?
Read Lyons and Guldi excerpts. Thinking with Lyons, what are some of the broader implications of categorizing in this way?

Readings / other media:

FAO/UNESCO Soil Map of the World | FAO SOILS PORTAL | Food and Agriculture Organization of the United Nations

US National Soil Archive

Kristina Lyons, Vital Decomposition, Introduction

Michel Rolph Trouillot, Silencing the Past, Chapter 1. 'The Power in the Story'.

Bowker and Star, Sorting Things Out (excerpt)

Guldi, The Long Land War Ch. 5 – 'On Failing to Make a Map in Time'.

Bruno Latour, Circulating Reference

See also the links in the Soil Collections section of Shannon Mattern's 'Library Field' Syllabus

Day 2: Archives conspicuously without soils

Some archives are about soil, and intentionally index soil within archival frameworks. Some archives go to elaborate lengths to remove soil – we might say, following Michel-Rolph Trouillot, to silence soil – and we can see an example of this in the modern herbarium. Plant collectors laboriously disentangle plants from soil, using chemical washes and dehydrators in order to mount plants flat on a piece of paper. These herbarium sheets are then archived and organised according to evolutionary relationships. Soil information is very rarely included, except for in one herbarium, which is called the 'Codex Badiano'. This is the oldest extant medicinal text in the Americas, and its illustrations

of plants always include roots and soil. Clearly, its authors thought that soil was an important part of how plants should be described. In this lesson, we will consider what the silencing of soil in botanical archives has meant historically, and what it might teach us about the metaphysics of our contemporary empirical practices.



Activity

Let's follow Michel-Rolph Trouillot's four moments through the botanical garden to examine the silences that may or may not be produced.

Step 1

Trouillot's first moment is the moment a plant is collected. In this moment, the plant is pulled out of the ground, meticulously disentangled from the soil, and made into a 'type'. Before the plant was pulled from the ground it was just another plant in a world full of plants, but in the moment it is 'collected' it begins to stand in for the species in general. It is no longer specific to that piece of ground, to that soil or to its relationship with sun and shade in the place it grew. It is now a specimen. Let's examine this moment of fact creation in the Missouri Botanical Garden archive.

Go to the Missouri Botanical Garden archival database.

Find a plant in the Missouri Botanic Garden (any plant) that you would like to explore. For example, try searching: '*Fraxinus americana*'.

To find Trouillot's first moment for this plant, go to one of the plant's herbarium sheets and look at it closely. Who collected this plant? What was their name? Where did they collect it? Next, explore the links associated with the plant. How many different ways has this plant (and plant species) been entered into this database? As you explore the archive, you are now exploring Trouillot's second moment: the moment where the collected plant becomes part of a larger archive.

Step 2

Trouillot's third moment is when we start to do things with the archive. This is when narratives are constructed through the ability to retrieve information that the archive has made available to our thinking.

Directly next to the species name (maybe *Fraxinus americana*) of your plant in the Living Collections Management System there will be a small Missouri Botanical Garden Logo. Click on this small logo. This will take you to Tropicos, the MOBOT taxonomic database. What can you learn about the plant here? Does the plant have an 'author'? Who is this person? What other plants did they identify?

In the case of *Fraxinus americana*, the author is Linneaus himself, meaning that Linnaeus was the first one to name this species. You'll also find citation information for the text in which the species first appeared in writing. In this case, Tropicos tells us that *Fraxinus americana* was first identified by Linneaus in the book *Species Plantarum* (1753). Next to this citation you'll see some more small logos ... one of them has the letters 'BHL'. Click on this link, which will take you to the Biodiversity Heritage Library site.

Here, you will see a digital scan of the physical book from 1753 where *Fraxinus americana* first appeared.

Step 3

Trouillot's final step is the making of history. This is where many narratives are considered and analysed in order to construct an account of the past. In this case, the plant was collected in 1941 by Jonathan Wright as part of a collection he called 'Plants of Indian Territory'. Could we think of this plant's history as exclusively taxonomic, or might we think of the legacy of settler colonialism as an important dimension of this plant's 'territory'?

In the final step, you will research and write your own history of this plant. Consider what aspects of this plant's history may have been left out of its presentation in the herbarium – for instance, more about the person who collected it and the circumstances under which they did so; or alternative contexts in which that plant may have gained meaning (e.g. within Indigenous knowledge systems). Drawing on this research, write a blurb for this plant that 'includes the soil'– that is, situates it within a political, historical, ecological and social context.

Readings / other media:

Robin Wall Kimmerer, 'Hearing the Language of Trees'.

Seth Denizen, 'Where is the Earth in the Herbarium'

Or watch: Seth Denizen: Where is the Earth in the herbarium? (Earth Writing, Faculty Workshop)

Camile Dungy, 'From Dirt'.

Maura Flannery, 'How to Read an Herbarium Specimen'.

Day 3: Soil as an archive of historical relations

Soils are created by many forms of circulation. Understanding soils and their mineral substrates, such as arsenic ore, help us tell histories of racism, empire, capitalism and landscape. Learn how chemical-intensive agriculture emerged, in part, through the social relations with and within soils.



Activity 1

Read Jayson Porter's essay in the essays section and consider it alongside the Ring of Fire map.

Activity 2

First, use EJScreen to see how close you live to a superfund site, hazardous waste or other toxins. Second, explore the 'Socioeconomic Indicators' tab to see how these indicators overlap toxic regions. Lastly, explore the 'Health Disparities' tab to further investigate the connections between toxicity, socioeconomic difference and health disparities.
Read about counter mapping:
– 'Counter Mapping', Emergence Magazine
– 'Counter-mapping: Cartography that Lets the Powerless Speak', The Guardian

Discuss: If counter mapping challenges 'dominant power structures to further seemingly progressive goals', how might counter-archiving require alternate forms of Trouillot's 'moment of fact retrieval'? Do you think the essay does the work of counter-archiving, and if so, how? Consider the historical case of arsenic. How might a soil archive of toxicity challenge power, counter dominant narratives in society and provide hope for the future?

Additional readings / other media

Max Liboiron, 'Pollution is Colonialism, Introduction'.
Adam Romero, 'Economic Poisoning', (excerpt, A Letter to Rachel Carson)
Rachel Carson, Silent Spring

Day 4: Soil as a living archive of social relations



Different people have different ideas of what is desirable (in the soil), and static concepts of soil health, fertility and even toxicity do not necessarily help us understand how soil works for or against people. This lesson asks us to imagine soil as a 'living archive' – a collection of different components – that is shaped by social and economic relations and which then shapes social possibilities. Students will also create their own archives to show that we can also use them to produce the knowledge we want.

Activity

Explore the Thinking Through Soil project's representations of soil, discuss with questions.

Assignment: Your soil archive

All waste passes through the soil at some point. Garbage, construction debris, bodily fluids, pharmaceuticals, nuclear fuel rods and socks with holes in them will one day make a soil somewhere. For a visualisation of this, please take a look at Seth Denizen's soil images. In this exercise, consider a day and your waste during that time, including human waste and the waste of materials you used or interacted with – what have you added to the soil?

- Write a list of the materials you discarded (or excreted) during the day
- Think about where these materials came from as well as where they go
- Consider how your own practices are shaped by the economic and social context you are a part of and are manifested in the soil that will be created

Readings / other media

Nicola Twilley, 'Soil Archive'.

Seth Denizen, 'Soil Contains Social Relations (and It's Our External Gut): Thinking Through Soil'.

Vanessa Agard-Jones, 'Bodies in the System'.

Thinking Through Soil